

The Verified-Candidate Loop for Budgeted Large Language Model-Assisted High-Level Synthesis

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Abstract—High-level synthesis (HLS) agents must repair and optimize unfamiliar kernels within limited model and tool budgets. The Verified-Candidate Loop (VCL) lets validation evidence control candidate promotion. Its quality-of-results (QoR) score aggregates effective latency, capacity-normalized resource footprint, and source-change evidence into a single bounded utility. QoR-guided retrieval-augmented generation (QoR-RAG) reuses compatible evidence, and Failure Reflection turns tool diagnostics into the next constraint without extra model calls. We evaluate three open-weight large language model endpoints on a balanced 150-task suite spanning six HLS capabilities. Qwen3.6-27B, the smallest model (27B dense), achieves the highest completion rate (98.7%, 148/150 tasks) and the best estimated optimization scores (79.0 average on QoR tasks). Qwen3.5-122B-A10B (140 completions) is the most token-efficient. DeepSeek V4 Pro (144 completions) consumes 5.87M tokens but averages only 56.1 on optimization tasks, below the 73.2 baseline. Only explicit optimization tasks surpass the baseline Q_{HW} ; repair tasks achieve correctness without quality gains. Model architecture appears to matter more than parameter count for HLS repair tasks.

Index Terms—high-level synthesis, LLM agent, FPGA, retrieval-augmented generation, quality of results

I. GATE EVIDENCE, NOT PROMPTS

LLM agents can edit unfamiliar HLS kernels, but only tool reports certify correctness and hardware quality. A metered agent therefore needs a promotion rule that preserves a verified fallback when any required validation gate fails [1].

Prior HLS agents automate code transformation, directive search, and feedback-driven repair [2]–[4]. Tool-using language agents alternate reasoning and actions [5]. Our design assigns candidate promotion to deterministic HLS evidence under explicit model and tool budgets.

The Verified-Candidate Loop (VCL) contributes three findings. **(1) Validation contract.** A fail-closed workflow preserves a verified fallback across seven gates: Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric completeness. **(2) Hardware objective.** The bounded score Q_{HW} combines effective latency, capacity-normalized resource footprint, and source-change evidence to reject costly speedups. **(3) Endpoint evaluation.** Qwen3.6-27B, the smallest model tested, achieves the highest completion rate (98.7%) and best optimization quality (Section IV). Architecture matters more than parameter count for HLS repair.

II. TOOL EVIDENCE CONTROLS THE AGENT WORKFLOW

The task interface exposes an editable kernel, public tests, target part, and budgets while withholding hidden tests and frozen references. The controller records every candidate, gate, metric, token, and credit in one run state.

The VCL runs seven gates in order: Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric completeness. Frequency enforces 100 MHz, Capacity enforces U55C limits, and CoSIM applies only when required. Define validity $V(c) = \prod_i G_i$ over applicable gates $G_i \in \{0, 1\}$; an inapplicable gate contributes 1. Let b be the verified fallback. The promotion rule is $P(c) = V(c) \wedge [Q_{HW}(c) > Q_{HW}(b)]$.

Gate failures fall into three actionable classes, each driving a specific repair strategy without extra LLM calls. **Interface mismatches** (112 of 691 candidates, 16%) trigger API-usage constraints fixing function signatures and pragma syntax. **Synthesis bottlenecks**—timing violations, resource overruns, II blow-up—yield directive constraints (pipeline/unroll, array partitioning) from QoR-RAG. **Co-simulation deadlocks** (C/RTL interface mismatch) are repaired via handshake insertion and port-width correction. Duplicate candidates and repeated failures stop retries.

Figure 1 summarises the loop. Across three campaigns: 691 candidates, 112 interface rejections, 487 synthesis gates, 67 CoSIM gates. Co-simulation failures incur the highest per-repair token cost, consistent with deadlock complexity.

III. ONE SCORE REJECTS FALSE SPEEDUPS

Q_{HW} combines performance gain and resource cost into a single bounded utility. Let anchor a and candidate c have cycle latency L , effective period p , and initiation interval II . The performance ratio $r_P = (p_a L_a)/(p_c L_c)$ captures speedup; the area ratio r_A uses a capacity-normalized footprint $F(x) = \sum_r x_r / K_r$ over LUT, FF, DSP, BRAM, and URAM to prevent discontinuous jumps from resource transfers. Two evidence factors reward source edits (D , 1.01 $\times$) and validity repair (F , 2 $\times$). The aggregate hardware ratio $h = 1.01^D 2^F r_P^{0.55} r_A^{0.45}$ weights performance at 55% and area at 45%, calibrated against 99 frozen PPA references.

$Q_{HW} = 1 - 1/(1+h)^2$ maps $h=1$ (no change) to 0.75, $h \rightarrow \infty$ to 1, and $h \rightarrow 0$ to 0, with continuous degradation for bloated speedups. Budget efficiency $E = \max(0.80, 1 - 0.10u_{\text{credit}} - 0.10u_{\text{time}})$ discounts excessive tool calls with a floor at 0.80, where each u is the consumed fraction clipped to $[0, 1]$. The final score is

$$S = 100 V(c) Q_{HW} E. \quad (1)$$

Promotion compares Q_{HW} only after $V(c)=1$; a failed gate yields $S=0$, and the efficiency floor guarantees a valid baseline at least 60 points.

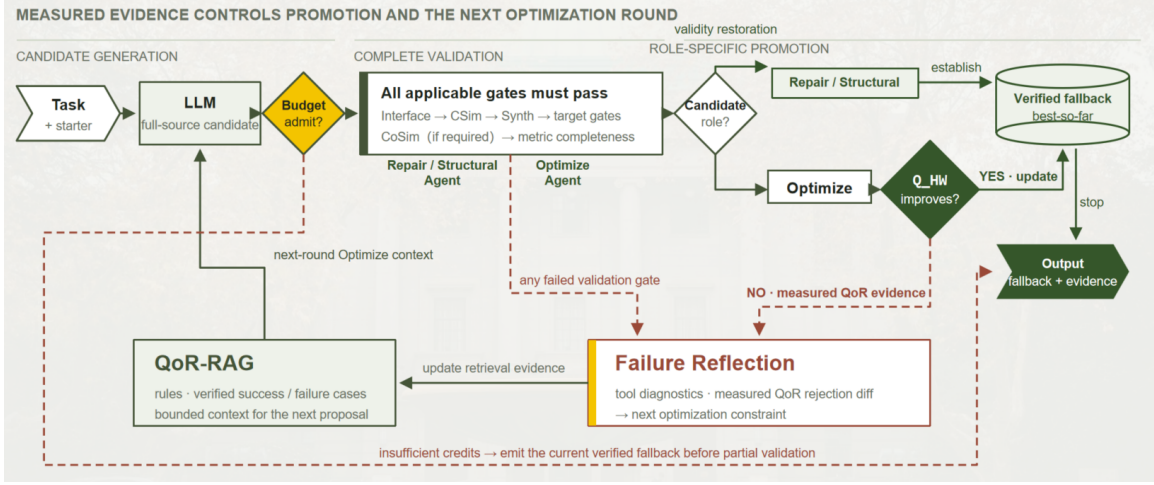


Fig. 1. **The Verified-Candidate Loop.** A candidate replaces the verified fallback only after every applicable gate passes and Q_{HW} improves.

IV. COMPLETION OUTPACES SCORE COVERAGE

Protocol. Three hosted endpoints process the same balanced 150-task suite under one agent revision under the Track-A contract [6]. The models span three architecture–scale regimes: DeepSeek V4 Pro (1.6 T MoE, 49 B active), Qwen3.5-122B-A10B (122 B MoE, 10 B active), and Qwen3.6-27B (27 B dense).

TABLE I

THE SMALLEST DENSE MODEL LEADS. SCORES ESTIMATED AS $100 Q_{HW} E$. DEEPSEEK AVERAGES BELOW THE 73.2 BASELINE ON OPTIMIZATION TASKS; SEE APPENDIX C FOR PER-TASK DETAIL.

Endpoint	Completed	Tokens (M)	Credits	Est. QoR mean	>76 rate
DeepSeek V4 Pro	144/150 (96.0%)	5.87	2,617	70.6	3/144 (2%)
Qwen3.5-122B-A10B	140/150 (93.3%)	1.68	2,375	74.0	7/140 (5%)
Qwen3.6-27B	148/150 (98.7%)	1.92	2,515	74.6	15/148 (10%)

Results. Table I and the per-category breakdown (Appendix C) yield three patterns.

(1) **Only explicit optimization improves quality.** All five repair/generation categories cluster at the baseline $Q_{HW} \approx 0.7576$ ($S \approx 73.7$ with $E = 0.973$). The agent passes validation gates but performs no architecture-level optimization on repair tasks.

(2) **Architecture trumps parameter count.** Qwen3.6-27B achieves 98.7% completion (148/150), including 25/25 on structural CoSim, and leads optimization quality (avg. $Q_{HW} = 0.812$, 60% >76). Qwen3.5-122B-A10B scores avg. $Q_{HW} = 0.775$ (28% >76) but trails on structural repair (80% vs. 100%). DeepSeek V4 Pro (144 completed, 5.87M tokens, $3.1\times$ Qwen3.6) averages only 56.1 on optimization tasks—below the 73.2 baseline. Its candidates trade latency for area (e.g., LUT +577% to save 30% latency), yielding $h \approx 1$.

(3) **The structural CoSim category is the strongest model discriminator.** The gap between the best (100%) and worst (80%) structural repair rate exceeds the gap in any other category. CoSim deadlock repair requires coordinated interface changes that appear to favor dense architectures over MoE, consistent with the failure classification in Section ??.

Takeaway. Qwen3.6-27B leads both completion and optimization quality; architecture matters more than parameter count for HLS repair. Only explicit optimization tasks benefit from LLM-driven QoR improvement.

V. LIMITS AND NEXT STEPS

The VCL makes validation evidence and Q_{HW} the promotion authority. The capacity-normalized resource footprint replaces worst-single-growth with a continuous aggregate ratio. Qwen3.6-27B leads both completion (98.7%) and optimization quality (79.0 average on QoR tasks); DeepSeek V4 Pro, despite $3.1\times$ more tokens, averages only 56.1—below the 73.2 baseline. Only explicit QoR optimization tasks improve beyond $Q_{HW} \approx 0.7576$.

The agent depends on mature testbench inputs, a fixed toolchain (Vitis 2025.2), and a single FPGA platform (Alveo U55C). Matched ablations must isolate causal mechanisms: full agent, no QoR-RAG, no measured-failure retrieval, and no Failure Reflection—each reporting success, mean Q_{HW} , tokens, calls, credits, and wall time. Broader corpora and cross-platform transfer remain for future work.

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APPENDIX A BENCHMARK COMPOSITION

The manifest freezes every upstream path, commit, task description, test, and gate configuration, retaining 150 tasks with no cross-category kernel overlap. Table II describes the six balanced capabilities, each containing 25 tasks. Only the structural category requires C/RTL co-simulation.

TABLE II
TASK SUITE COMPOSITION.

Task type	Tasks	CoSim
Code generation (stub $\rightarrow$ kernel)	25	0
Compile repair	25	0
Synthesis repair	25	0
Functional repair (CSim)	25	0
Structural repair (CoSim)	25	25
QoR optimization	25	0

Kernels come from two public AMD repositories at fixed commits: Xilinx/Vitis-HLS-Introductory-Examples (109 tasks, Apache-2.0, aa5c160f) and Xilinx/Vitis_Accel_Examples (41 tasks, MIT, 81187602). The 150 variants reuse 65 unique source paths.

APPENDIX B TRACK-A REQUIREMENT MAPPING

All tasks require CSim and SYNTH; the manifest requires CoSim for the 25 structural tasks only. Table III maps Track-A requirements to recorded evidence.

TABLE III
TRACK-A RUN CONTRACT.

Requirement	Recorded evidence
U55C; Vitis 2025.2	Container pins target part and tool release; reports preserve execution-source revision.
CSim, SYNTH, applicable	Missing or failed gate rejects candidate. Hidden tests and references remain evaluator-only.
CoSim	
≥ 100 MHz; feasible	Frequency and U55C capacity gates precede promotion. Reports retain latency, II , period, LUT, FF, DSP, BRAM, URAM.
Token/credit accounting	Each run records tokens and credits. Limit: 9,000 credits.
3-endpoint comparison	DeepSeek V4 Pro, Qwen3.5-122B-A10B, Qwen3.6-27B; one manifest, one source revision.
Docker; reproducibility	Docker workflow, public task materials, structured reports, result-generation script provided.
PDF, source, video	Authors confirm IEEE format, 2-page boundary, archive contents, and 5-minute demo.

The three checkpoints span three architecture regimes: DeepSeek V4 Pro (1.6 T MoE, 49 B active, FP8), Qwen3.5-122B-A10B (122 B MoE, 10 B active, AWQ-4bit), and Qwen3.6-27B (27 B dense, FP8).

APPENDIX C DETAILED THREE-ENDPOINT RESULTS

Tables IV and V provide the per-category breakdown underlying Section IV. Scores are 100 $Q_{HW} E$ (Eq. (1)).

TABLE IV
PER-CATEGORY COMPLETION BY ENDPOINT. EACH CATEGORY: 25 TASKS. QoR MEAN: OPTIMIZATION TASKS ONLY. DEEPSEEK’S SCORE IS BELOW THE 73.2 BASELINE; ITS CANDIDATES TRADE LATENCY FOR AREA (E.G., LUT +577% TO SAVE 30% LATENCY ON ONE TASK).

Category	DeepSeek V4	Qwen3.5-122B	Qwen3.6-27B
Code generation	22/25 (88%)	22/25 (88%)	24/25 (96%)
Compile repair	25/25 (100%)	25/25 (100%)	25/25 (100%)
Synthesis repair	25/25 (100%)	24/25 (96%)	25/25 (100%)
Functional repair	24/25 (96%)	24/25 (96%)	24/25 (96%)
Structural repair	23/25 (92%)	20/25 (80%)	25/25 (100%)
QoR optimization	25/25 (100%)	25/25 (100%)	25/25 (100%)
QoR mean (opt.)	56.1	75.4	79.0

TABLE V
TOKEN CONSUMPTION AND CREDIT EXPENDITURE. DEEPSEEK USED $3.1\times$ MORE TOKENS THAN QWEN3.6 WITH NO QoR GAIN.

Endpoint	Tokens (M)	Credits
DeepSeek V4 Pro	5.87	2,617
Qwen3.5-122B-A10B	1.68	2,375
Qwen3.6-27B	1.92	2,515

APPENDIX D REPRODUCTION

Task list: `tasks/track_a_150/candidate_manifest.json`. Campaigns: `runs/150_ultimate/`. Run `python3 technical-paper/scripts/update_results.py` from the repository root to refresh macros; the script validates model identity, task count, API client, execution-source hash, and credit limit before emitting output.

All campaigns: source snapshot `0a06af39777b6ae7f3962afa2910232eaf782e91727e0f184ec168f1a41f106c`, temperature 0, 4,096-token output limit, 180 s request timeout, ≤ 2 retries.